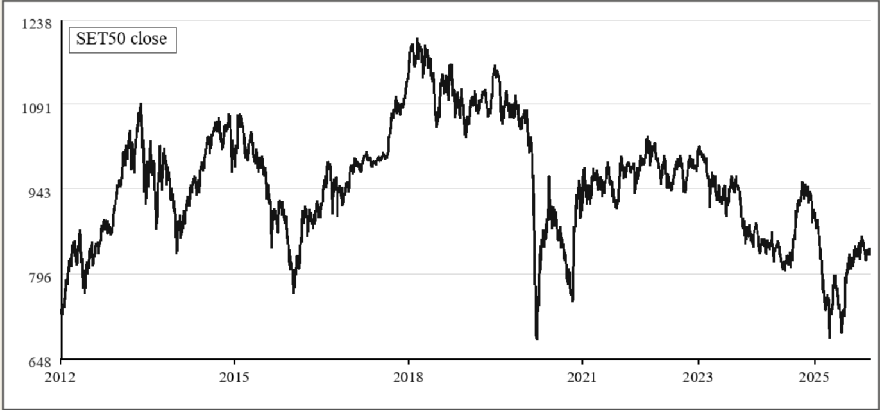


Input data

MARKET SERIES

SET50 daily index

2012-2025 | 116 TA features | t+1 target



All numerical features are reconstructed using only information available by the close of day t.

FINANCIAL-NEWS MODALITY

Dated news to out-of-sample sentiment

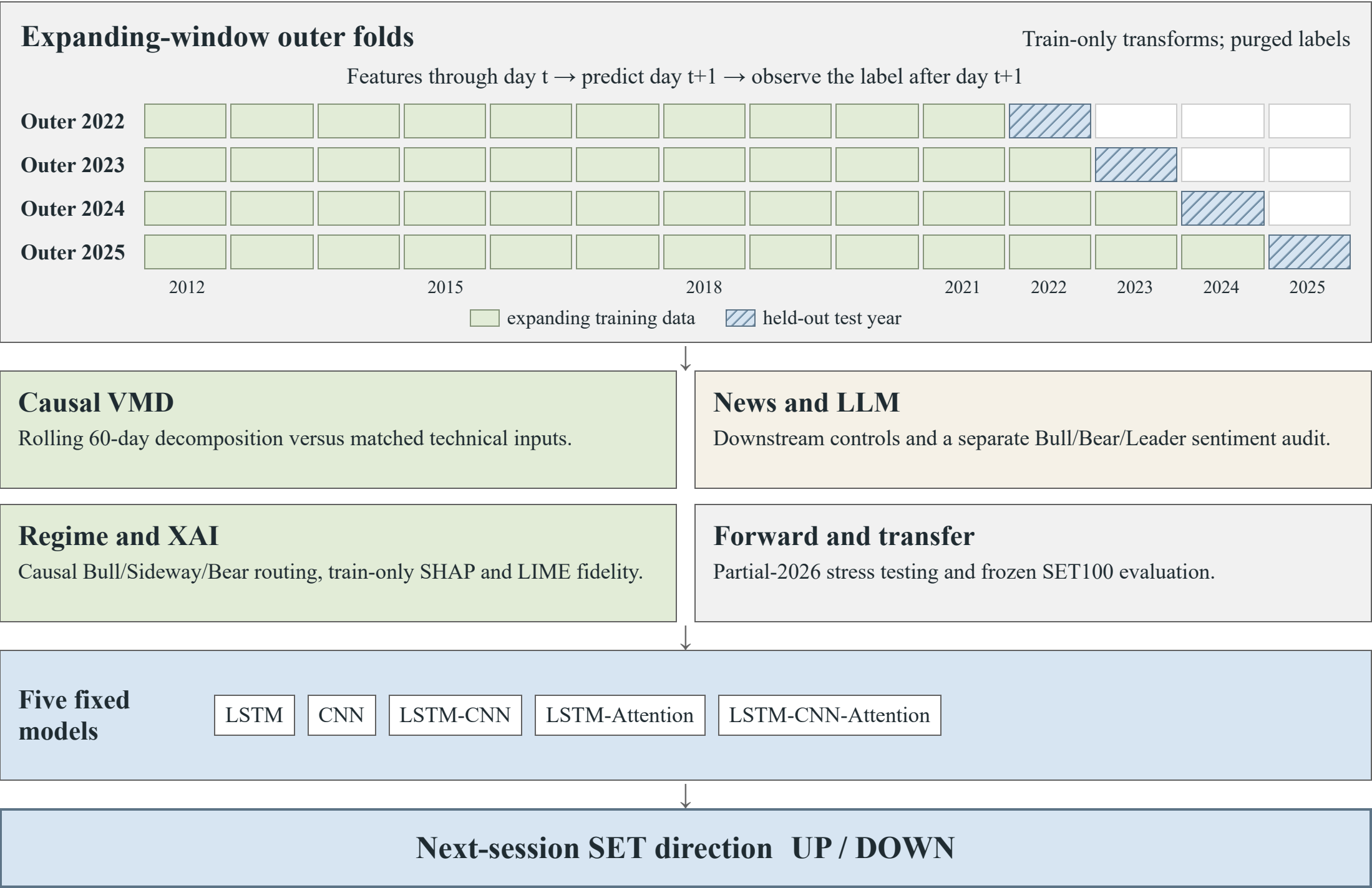
Dated articles

Date, ticker and relevance filter

Expanding OOS sentiment

Sentiment is generated without using labels from the subsequent forecasting period.

Point-in-time reliability evaluation



Main findings

Best mean balanced accuracy

LSTM-CNN-Attention; four held-out years and five seeds.

53.64%

CNN with Regime-SHAP

Largest selective forecasting gain.

+1.46 pp

Leader sentiment

Versus equal-call and near-cost controls; intrinsic endpoint.

+5.93 / +6.00 pp

LIME fidelity stress test

Share of repeated explanations classified as low fidelity.

71.83%

Robustness results

VMD effect: -0.60 to +0.35 percentage points.

Predicted news: no consistent adjusted forecasting gain.

Partial-2026 stress test: one-sided collapse in several objectives.

SET100 transfer: lower mean balanced accuracy for all five models.

Observed improvements were architecture- and endpoint-dependent rather than universal.